

Predicting APS Failures in Scania Trucks

A cost-sensitive comparison of machine learning classifiers on the IDA2016 APS Failure dataset

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FINAL BEST MODEL
XGBoost at threshold 0.35

Cost: 9,900 Recall: 0.987 FN: 5

Abstract & Objective

This project predicts whether a heavy truck record is related to an Air Pressure System (APS) failure or to another component. The task is difficult because the dataset is highly imbalanced and false negatives are much more costly than false positives.

Goal: compare six machine learning classifiers and choose the model that minimizes operational cost, not just accuracy.

60,000

training records

16,000

test records

~170

numeric features

50x

FN vs FP cost

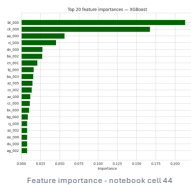
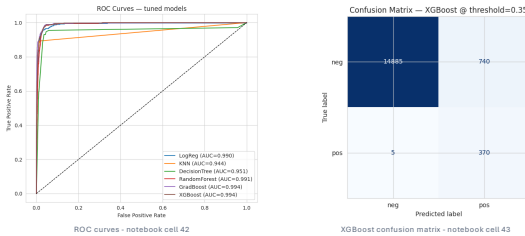
Methodology



Six classifiers were trained: Logistic Regression, KNN, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. Random Forest and XGBoost satisfy the required ensemble methods.

Evaluation focused on recall and total cost because missing a real APS failure is expensive. The custom cost function was: Total Cost = $10 \times \text{FP} + 500 \times \text{FN}$.

Model Diagnostics



At threshold 0.35, XGBoost produced TN=14,885, FP=740, FN=5, TP=370.

Key EDA Visualizations



EDA takeaways:

- Target is severely imbalanced: about 98.3% neg vs. 1.7% pos.
- Several columns have very high missingness; median imputation is necessary.
- Many features are heavy right-skewed, so scaling matters for LogReg/KNN.
- Boxplots show visible separation between classes, meaning learnable signal exists.

Results: Model Comparison

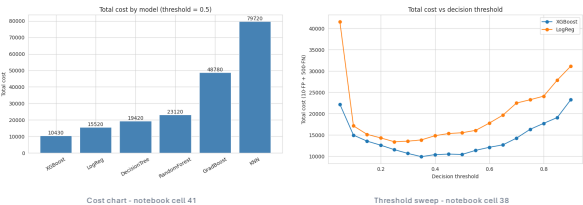
Model	Precision	Recall	ROC-AUC	FP	FN	Cost @ 0.5	Best Threshold	Tuned Cost
XGBoost	0.382	0.976	0.994	503	9	10,430	0.35	9,900
LogReg	0.499	0.936	0.990	352	24	15,520	0.25	13,410
DecisionTree	0.352	0.931	0.951	642	26	19,420	N/A	N/A
RandomForest	0.518	0.893	0.991	312	40	23,120	N/A	N/A
GradBoost	0.908	0.741	0.994	28	97	48,780	N/A	N/A
KNN	0.908	0.576	0.944	22	159	79,720	N/A	N/A

Main comparison table - notebook cell 40

Main result:

- Tuned XGBoost had the best default-threshold cost: 10,430.
- Threshold tuning lowered XGBoost cost further to 9,900.
- Accuracy alone was not reliable because a majority-class model could look strong while missing APS failures.

Cost-Focused Findings



Threshold tuning shows why 0.5 was not optimal. Lowering XGBoost to 0.35 increased failure detection and produced the lowest total cost.

Conclusion & Future Work

Conclusion:

- XGBoost best handled skewed, high-dimensional data and class imbalance after tuning.
- The final choice was driven by operational cost, not accuracy alone.
- The best model missed only 5 APS failures after threshold tuning.

Future work:

- Apply stacking with Random Forest, XGBoost, Gradient Boosting, and Logistic Regression.
- Re-run the pipeline after dropping columns above 70% missingness.

References & Source Notes

https://github.com/AyhamAbbad/SP26-CSCI460_Aps_Failure.git